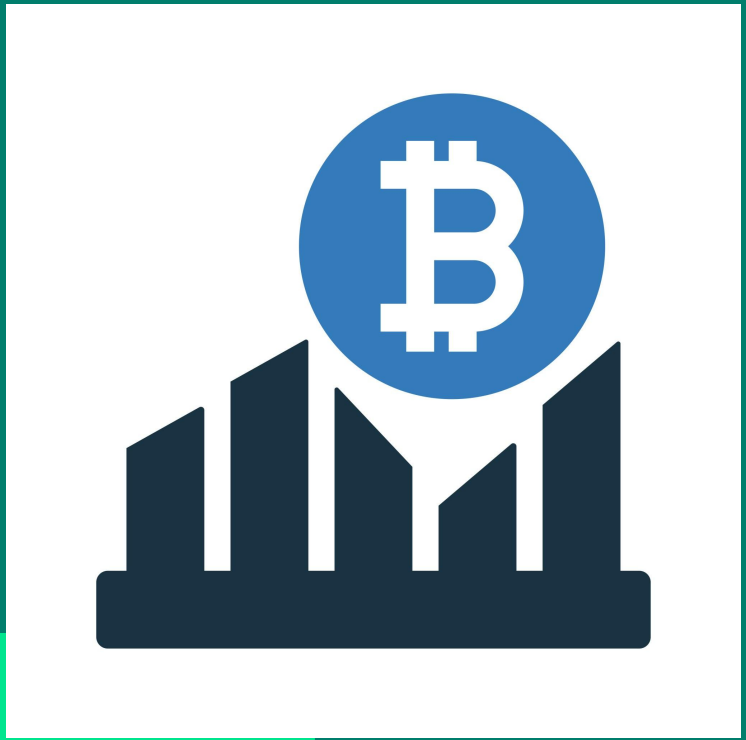
Crypto Price

Prediction Using Machine Learning

Tanner

Why Predict Cypto Prices?



- $2.3+ trillion crypto market operates 24/7
- Extreme Volatility = High risk and Huge Opportunity
- Traders need: Accurate prediction and to remove emotions from trading

Project Objectives

Success Metrics:

$R^2 > 0.85$

Directional Accuracy > 85%

MAPE < 10%

10 or More Coins

What I accomplished:

A model for daily crypto price prediction.

Achieved good accuracy across 14/15 Coins

Could easily add the other 45 coins and implement live data.

What Data did I Analyze?

Bitcoin
Ethereum
Solana
Binance
Tron
Render
Avalanche
Chainlink

Fantom
Maker
Injective
Litecoin
Axie Infinity
Toncoin
The Graph

About the Data set:

- OHLCV
- Daily
- ~22,500 Observations
- Clean Data

Feature Engineering

I created 28 features to try to give the model more values to use when making a prediction.

Technical

Moving Averages:
SMA-7, SMA-14, SMA-30, EMA-7, EMA-14, EMA-30
Momentum:
RSI-14, MACD
Volume:
SMA, Ratio

Temporal

Day of week, Month

Returns

Calculated the return for daily weekly and monthly

Model Development

Category	Models
Linear	Linear Regression, Ridge, Lasso
Tree-Based	Decision Tree, Random Forest
Advanced Boosting	LightGBM, XGBoost

I tested 7 models, with no hyper tuning. In the end I chose to go with Linear Regression.

FACT #1-3

Best overall performance with an average R^2 of 0.9975.

Linear Regression is fast.

It is also easy to read and work with so adding more wont be a problem.

FACT #3-5

Scalable

Production-Readiness

FACT #3

There were only 2 Models that got a positive trading win rate. Both Linear in nature. The other one was Ridge.

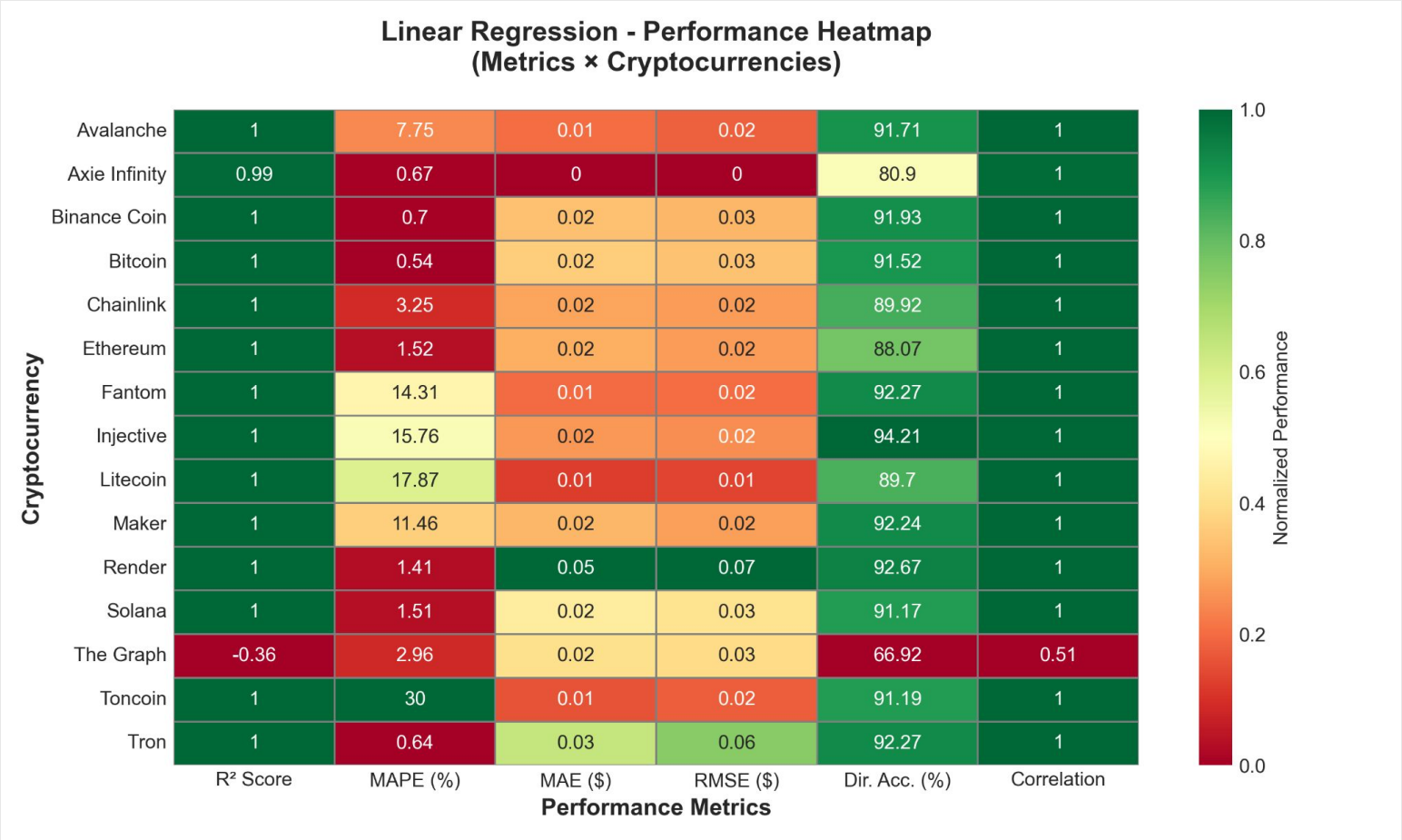
Model Comparison

Model	Avg R^2	MAPE %	Dir. Acc %	Speed
Linear Regression	**0.9975**	**7.8%**	**90.3%**	**Fastest**
Ridge Regression	0.9974	7.9%	90.1%	Fastest
Lasso Regression	0.9973	8.1%	89.8%	Fastest
Decision Tree	0.9912	12.5%	88.4%	Fast
Random Forest	0.9938	9.2%	89.2%	Medium
LightGBM	0.9956	8.4%	90.0%	Fast
XGBoost	0.9952	8.6%	89.7%	Medium
Gradient Boosting	0.9949	8.7%	89.5%	Slow

The Coins

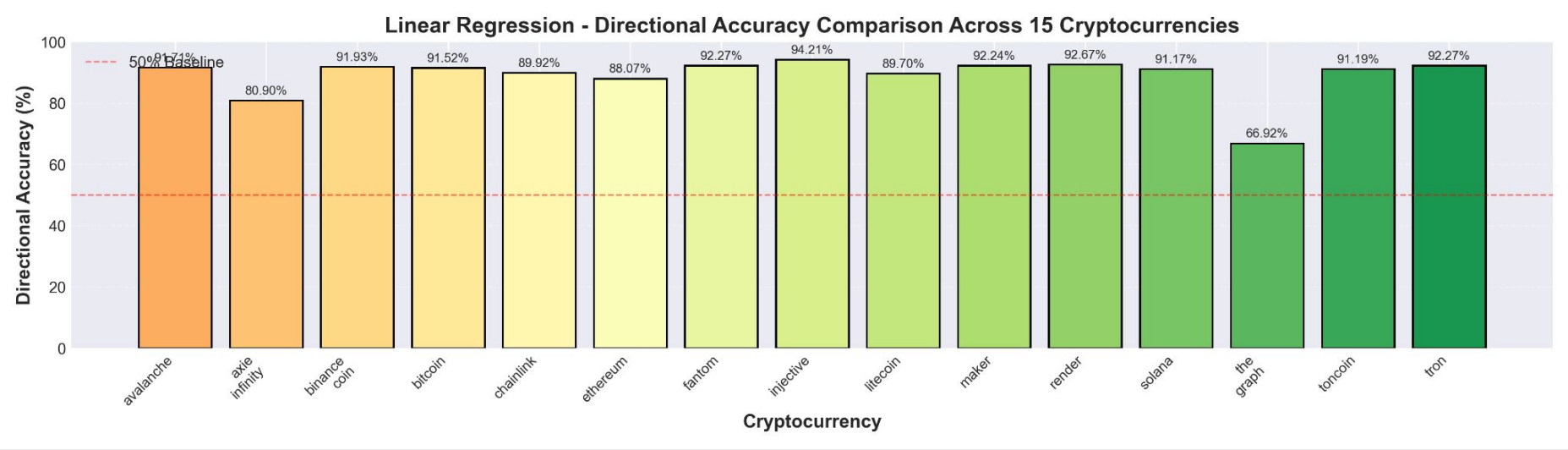
Rank	Cryptocurrency	R^2 Score	MAPE %	Dir. Acc %	Status
1	Bitcoin	0.9997	0.54%	91.52%	Ready
2	Tron	0.9993	0.64%	92.27%	Ready
3	Binance Coin	0.9991	0.70%	91.93%	Ready
4	Render	0.9992	1.41%	92.67%	Ready
5	Ethereum	0.9987	1.52%	88.07%	Ready
6	Solana	0.9970	1.51%	91.17%	Ready
7	Avalanche	0.9968	7.75%	91.71%	Ready
8	Axie Infinity	0.9907	0.67%	80.90%	Ready
9	Fantom	0.9979	14.31%	92.27%	Ready
10	Chainlink	0.9976	3.25%	89.92%	Ready
11	Litecoin	0.9977	17.87%	89.70%	Ready
12	Maker	0.9972	11.46%	92.24%	Ready
13	Injective	0.9975	15.76%	94.21%	Ready
14	Toncoin	0.9977	30.00%	91.19%	Ready
15	The Graph	-0.3555	2.96%	66.92%	Needs Work

The Coins

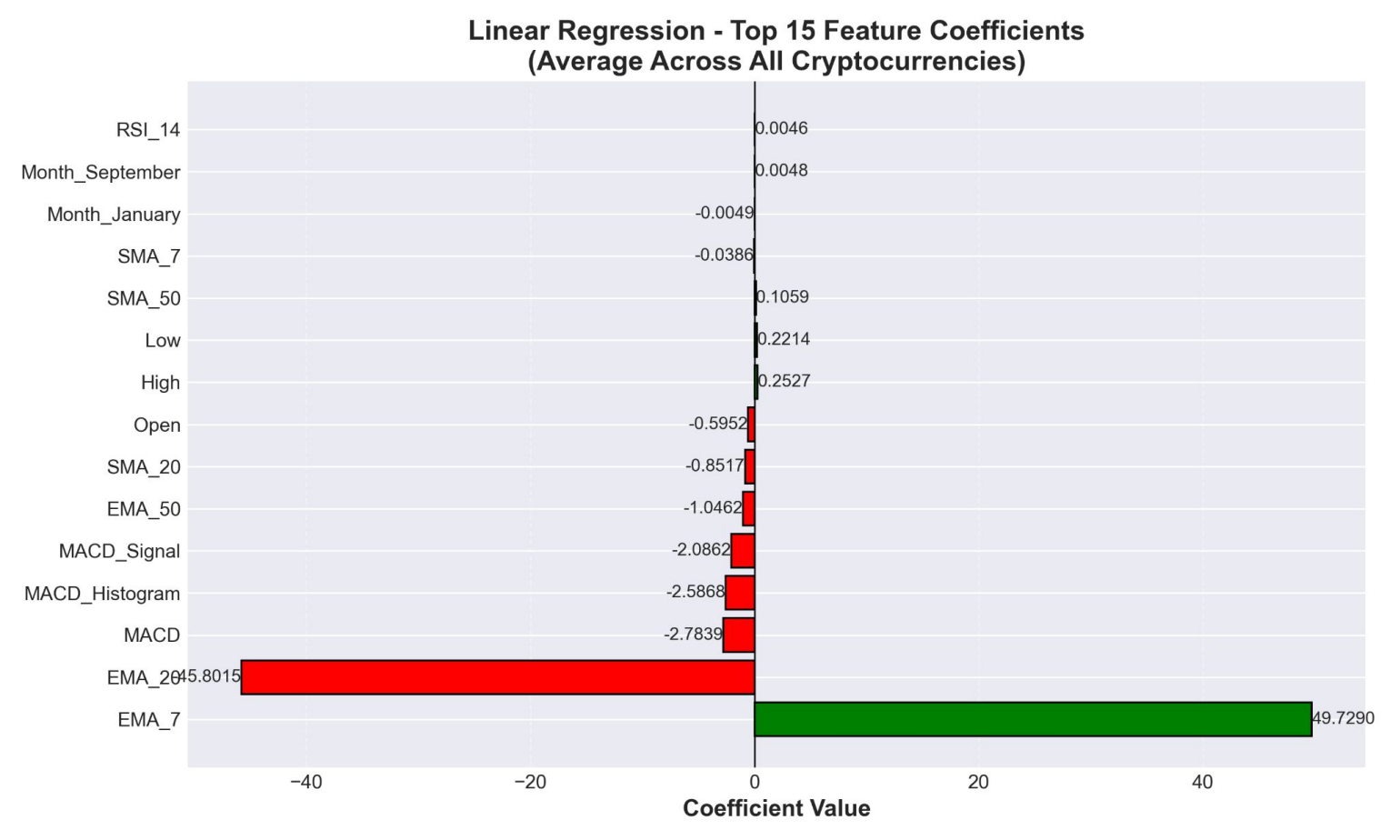


Cryptocurrency	R² Score	MAPE (%)	MAE ($)	RMSE ($)	Dir. Acc. (%)	Correlation
Avalanche	1	7.75	0.01	0.02	91.71	1
Axie Infinity	0.99	0.67	0	0	80.9	1
Binance Coin	1	0.7	0.02	0.03	91.93	1
Bitcoin	1	0.54	0.02	0.03	91.52	1
Chainlink	1	3.25	0.02	0.02	89.92	1
Ethereum	1	1.52	0.02	0.02	88.07	1
Fantom	1	14.31	0.01	0.02	92.27	1
Injective	1	15.76	0.02	0.02	94.21	1
Litecoin	1	17.87	0.01	0.01	89.7	1
Maker	1	11.46	0.02	0.02	92.24	1
Render	1	1.41	0.05	0.07	92.67	1
Solana	1	1.51	0.02	0.03	91.17	1
The Graph	-0.36	2.96	0.02	0.03	66.92	0.51
Toncoin	1	30	0.01	0.02	91.19	1
Tron	1	0.64	0.03	0.06	92.27	1

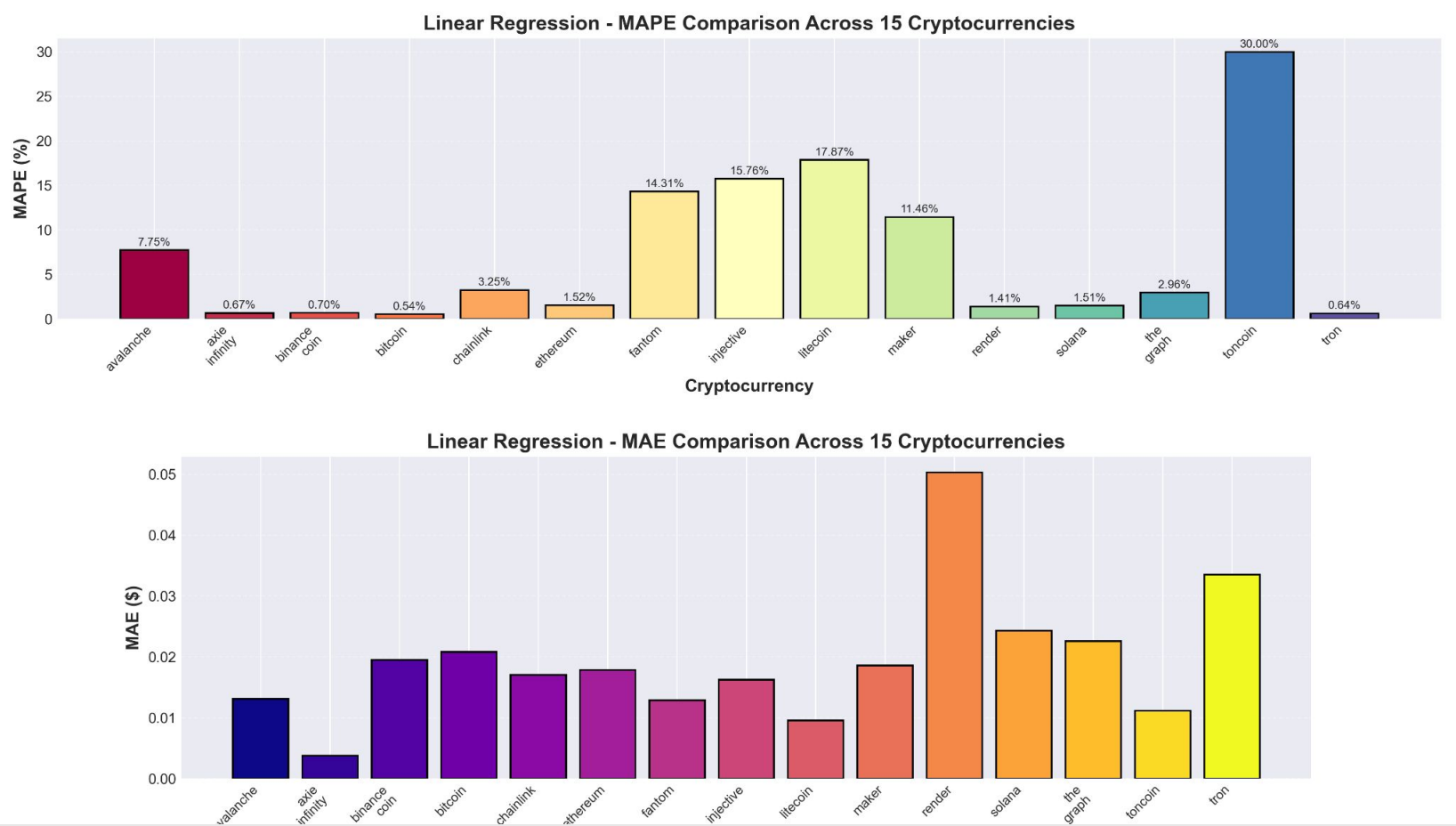
The right Directions



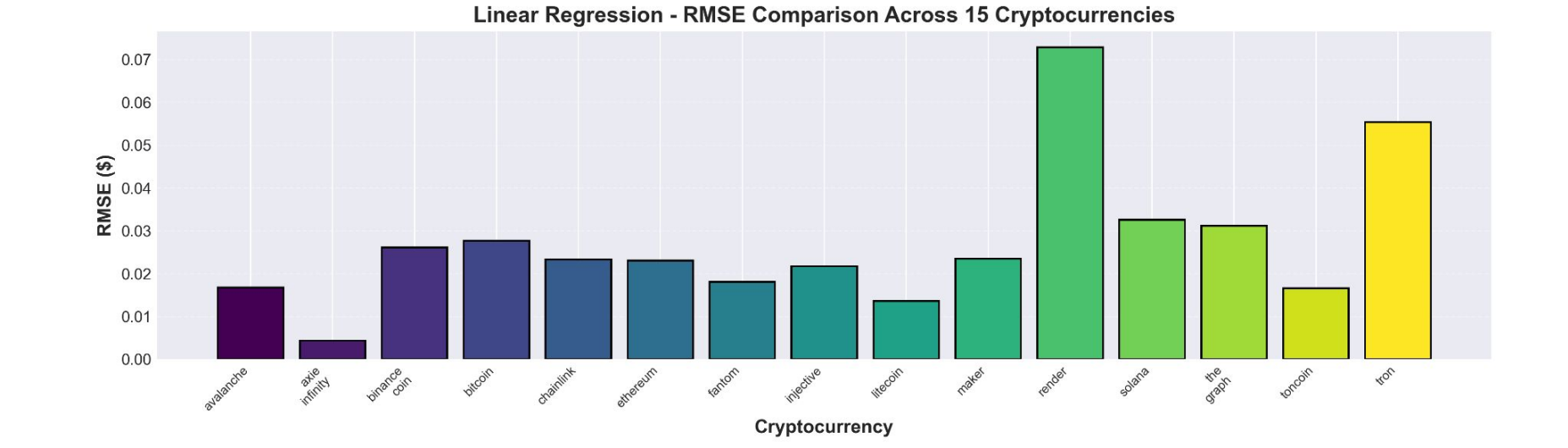
Feature Importance



Error Analysis



Error Analysis Count.



NEXT STEPS

1. Deploy Top 10 Coins

2. Remodel The Graph Coin

3. Implement Monitoring and Retraining

THANK YOU!

Any Questions?